BIMM 143 Lab 05

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Section 6 : Creating Scatterplots

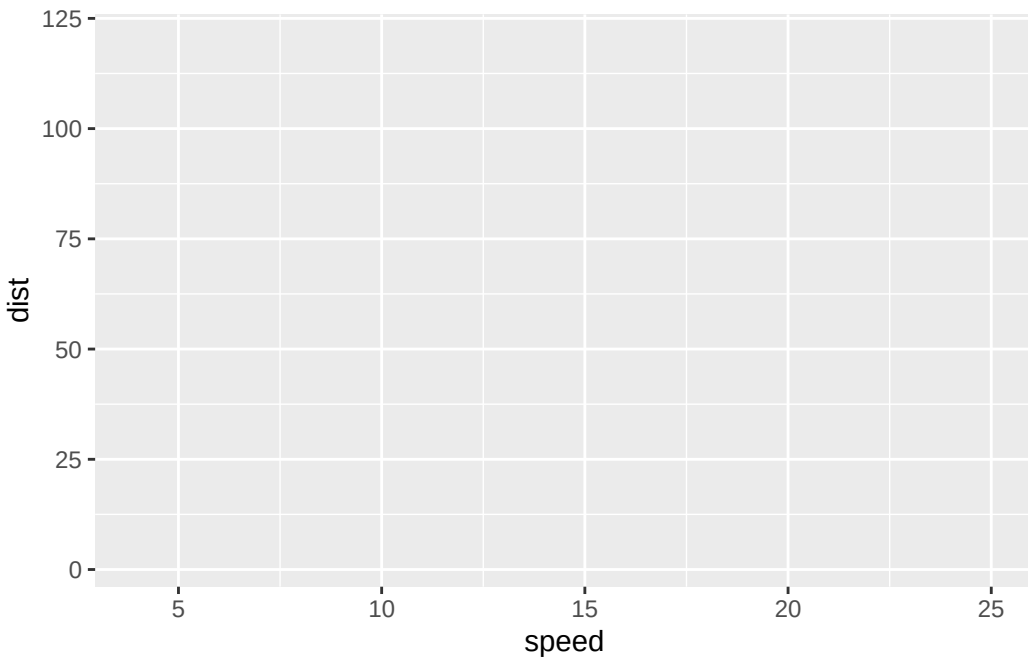
Part 1: Making a plot and specifying a data set

```
library(ggplot2)
ggplot(cars)
```

This makes the plane upon which our ggplot graph will go.

Part 2: Adding aesthetic mapping

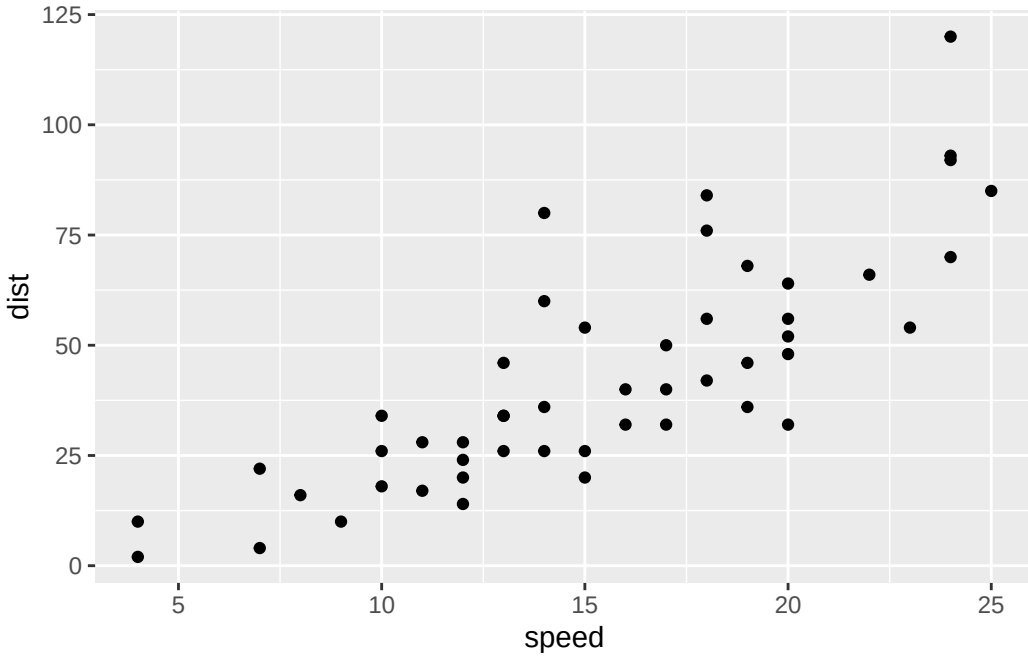
```r
ggplot(cars) +
  aes(x = speed, y = dist)
```



The aesthetic mapping creates the axes and lables them according to the column names unless otherwise specified.

Part 3: Adding geometry

```r
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point()
```



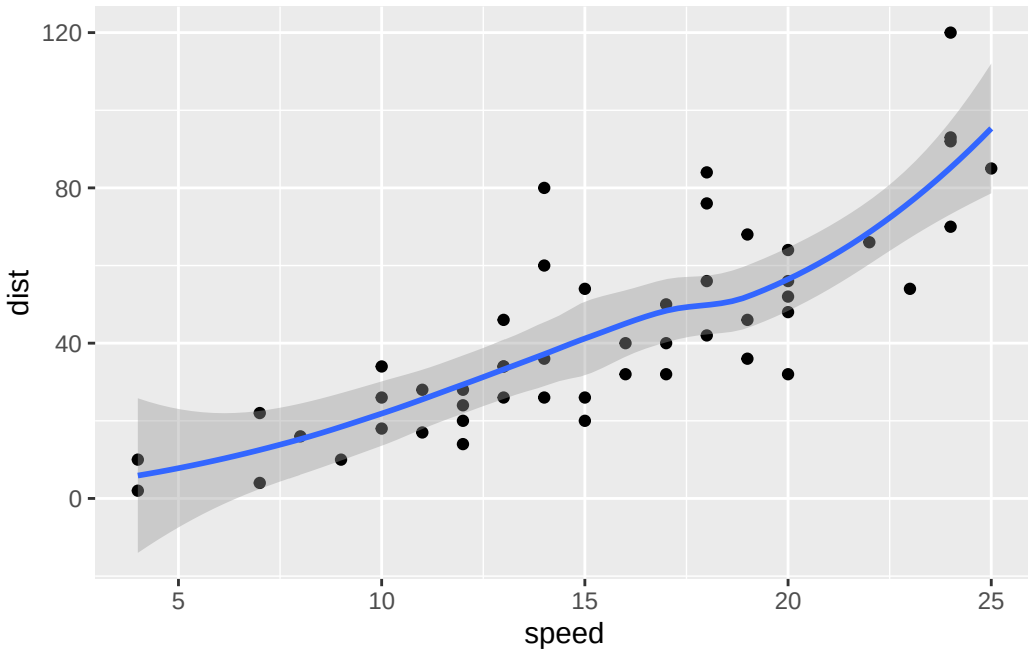
The geom_point() function adds a scatterplot on the geometry layer. Since we haven't specified any other details, R uses the default settings.

Q: Which geometric layer should be used to create scatter plots in ggplot2? : geom_point

Q: In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the geom_smooth() function?:

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

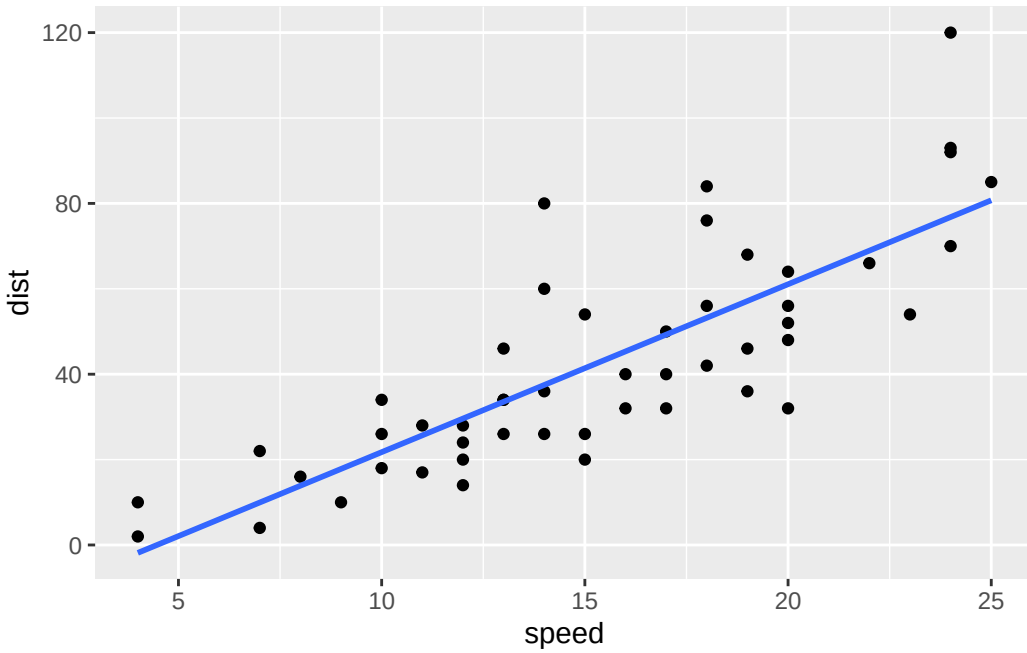


Here, geom_smooth() just adds a trendline that best fits the data, along with a shaded area that represents the standard error of the data.

Q: Argue with geom_smooth() to add a straight line from a linear model without the shaded standard error region?

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = lm, se = FALSE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

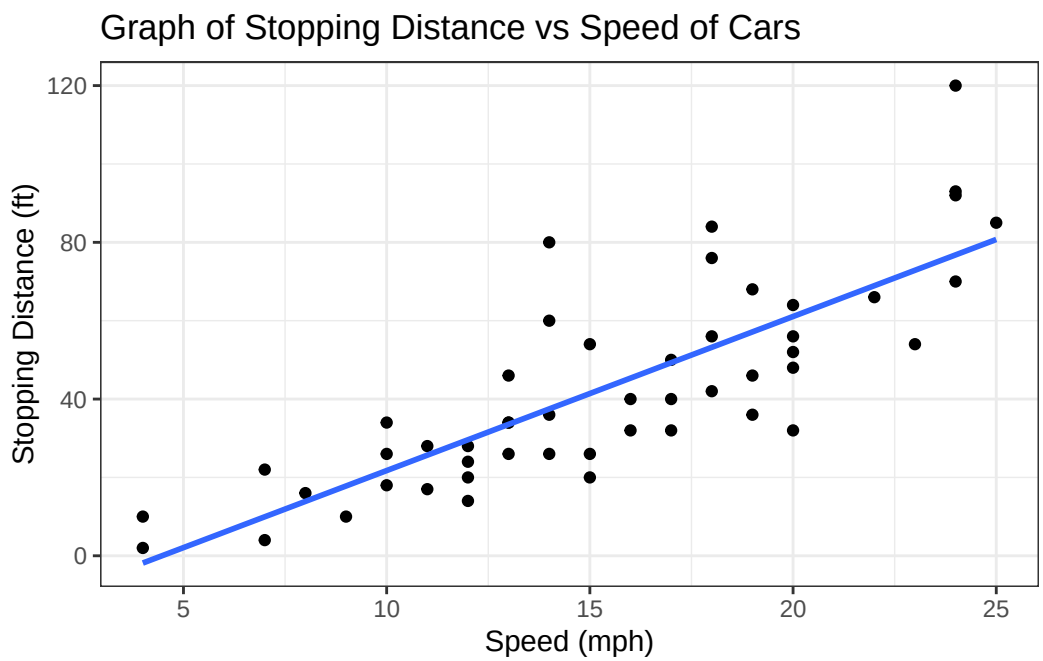


Inside the geom_smooth() argument, the method = lm code defines that the method used to create the smooth line graph should be a **l**inear **m**odel. The se argument defines whether or not the standard error appears shaded in grey on the graph. Here, because I have coded se = FALSE, it removes the shaded grey standard error.

Q. Can you finish this plot by adding various label annotations with the labs() function and changing the plot look to a more conservative "black & white" theme by adding the theme_bw() function:

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = lm, se = FALSE) +
  labs(
    title = "Graph of Stopping Distance vs Speed of Cars",
       x = "Speed (mph)",
       y = "Stopping Distance (ft)") +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Here, the Title argument under the labs() function adds the title, and the x and y arguments add the x and y axes labels respectively.

Part 4: Adding more plot aesthetics

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Q. Use the nrow() function to find out how many genes are in this dataset. What is your answer?:

```r
nrow(genes)
```

```
[1] 5196
```

There are 5196 rows in this dataset.

Q. Use the colnames() function and the ncol() function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```r
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```r
ncol(genes)
```

```
[1] 4
```

There are four columns and their names are "Gene", "Condition 1", "Condition 2", and "State".

Q. Use the table() function on the State column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

```r
table(genes$State)
```

```
      down unchanging         up 
        72       4997        127 
```

There are 127 upregulated genes.

Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

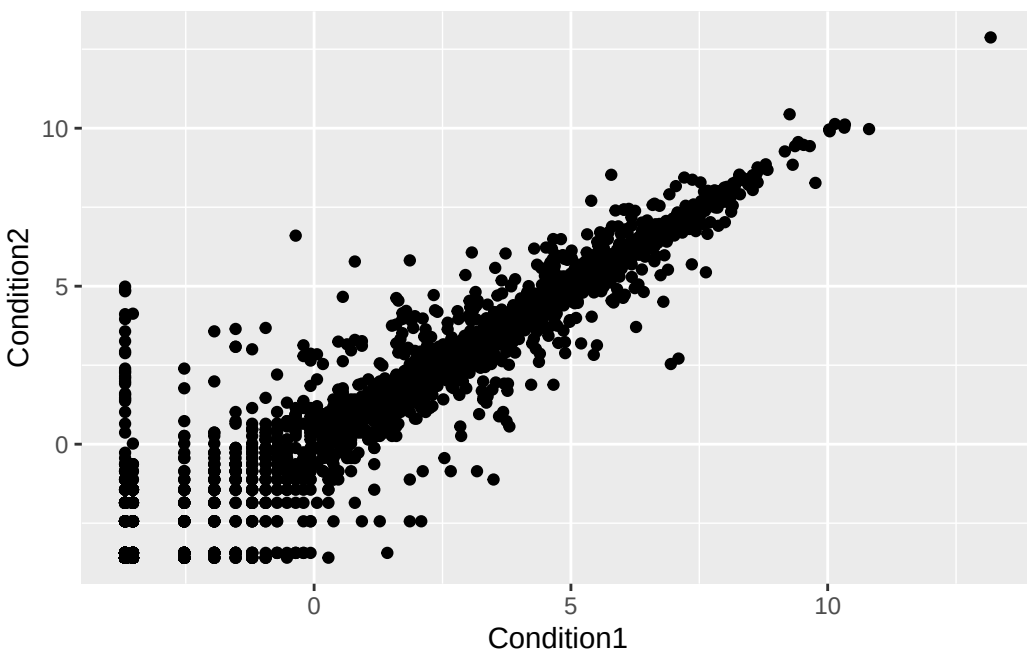
```r
fractions <- round((table(genes$State)/nrow(genes)) * 100, 2)
fractions
```

```
      down unchanging         up
      1.39      96.17       2.44
```

Here, I divide the whole table by the number of genes to obtain the fraction of each, and then multiple by 100 to obtain a percentage and then round to 2 decimal places as the question requires. The fraction of upregulated genes is 2.44%.

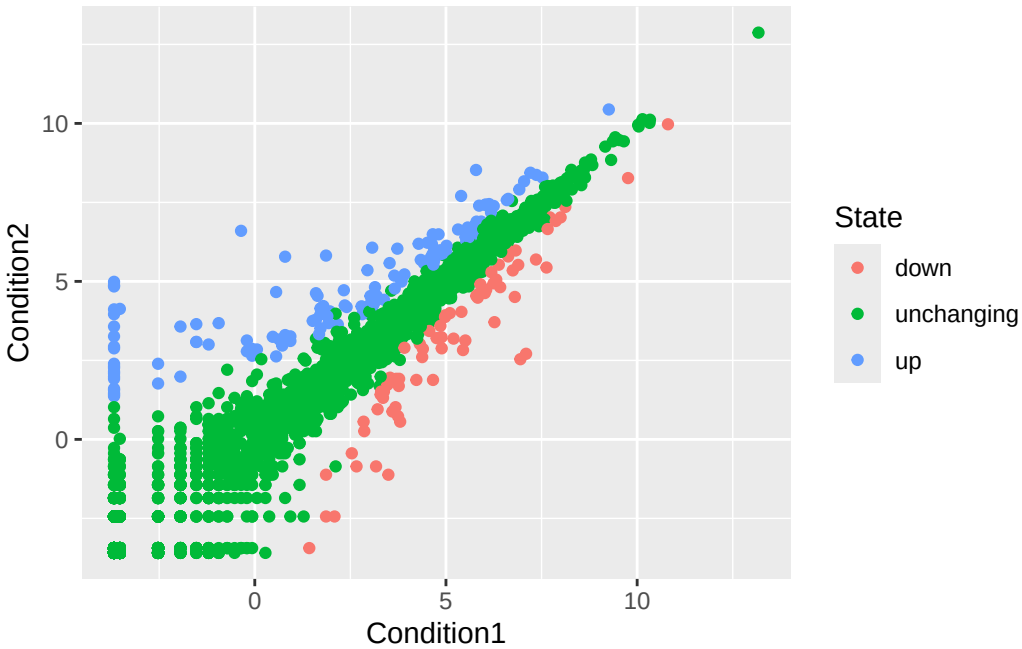
Making a scatterplot with the Genes dataset:

```
ggplot(genes,
       aes(Condition1, Condition2)) +
  geom_point()
```



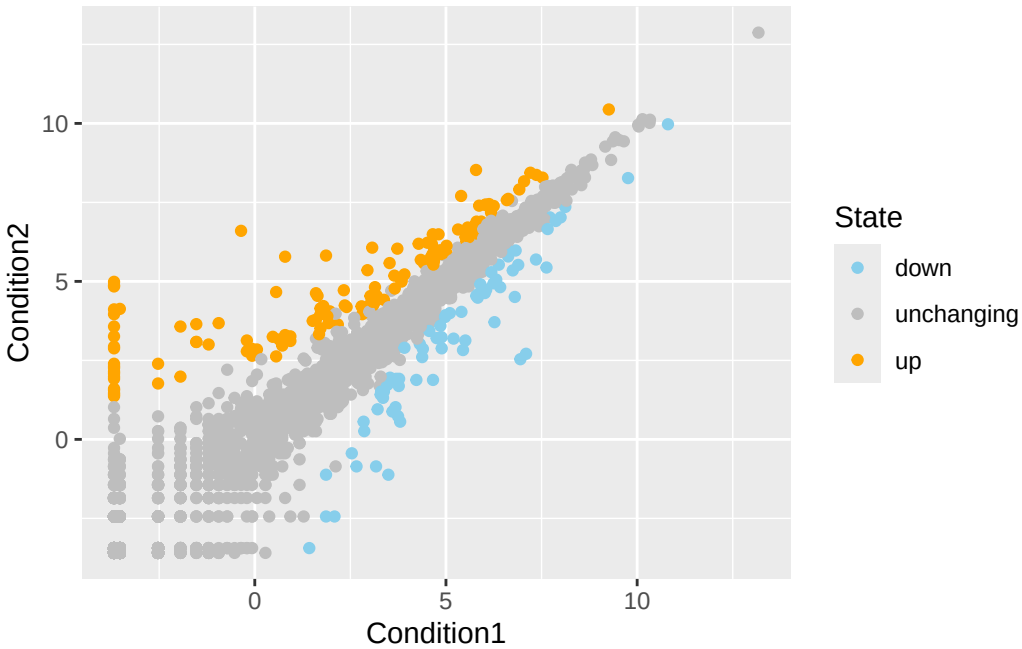
Colouring by State

```
plot <- ggplot(genes,
               aes(Condition1, Condition2, color = State)) +
  geom_point()
plot
```



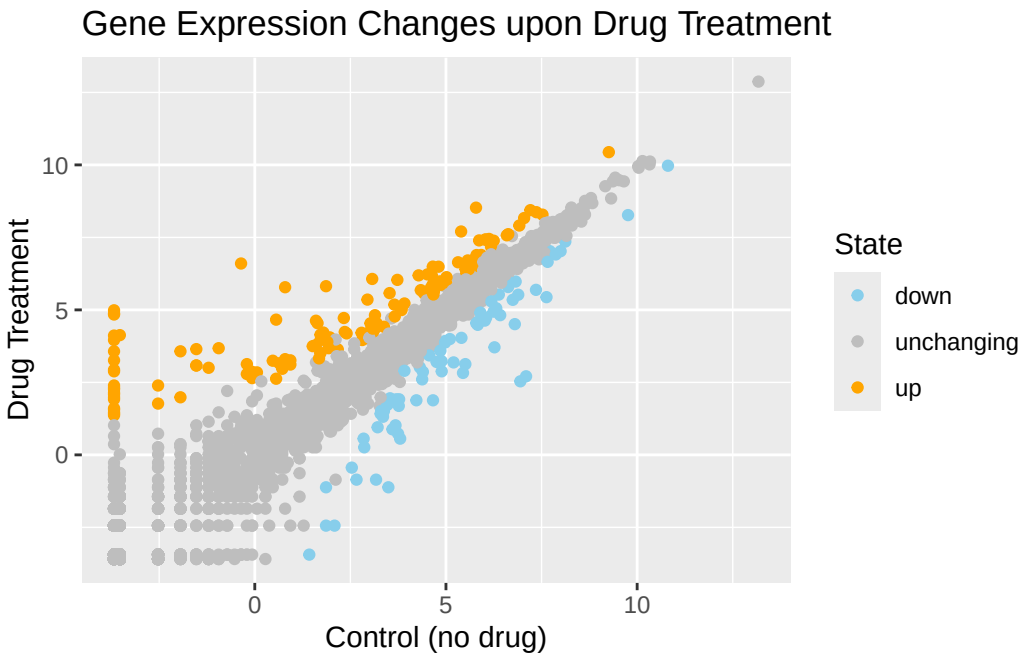
Adding a manual color scale

```
plot_colors <- plot +
  scale_colour_manual(
    values = c("skyblue", "grey", "orange"))
plot_colors
```



Adding labels:

```
plot_labels <- plot_colors +
  labs(
    title = "Gene Expression Changes upon Drug Treatment",
    x = "Control (no drug)",
    y = "Drug Treatment")
plot_labels
```


Gene Expression Changes upon Drug Treatment
Drug Treatment
10
5
0
0
5
10
Control (no drug)
State
down
unchanging
up